Lecture 8: Magnetohydrodynamics

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In this lecture, we will discuss a particularly important fluid model known as magnetohydrodynamics (MHD). In essence, MHD treats magnetized plasma as single conducting fluid and describes low-frequency large-scale phenomena.[1] There are many variations of MHD, and this lecture is not intended to be comprehensive. We will consider nonly the most basic ones: resistive MHD and, as a special case, ideal MHD.

1 The model

1.1 Assumptions and basic estimates

Some versions of MHD can be derived as a corollary to the Braginskii equations under additional assumptions. However, this approach leads to models that are too constrained for typical plasma-physics applications. In particular, MHD is widely used to model fusion plasmas, where the particle mean free path is in the km range and, therefore, Braginskii's key assumption $\lambda_{\mathrm{mfp}}/L \ll 1$ is not satisfied. Thus, we will revert to the general fluid equations that we derived earlier and try to build a general theory from scratch under new assumptions.

We assume nonrelativistic electron–ion plasma, with one type of singly charged ions, and

$$\epsilon \doteq \frac{\rho_i}{L} \sim \frac{\omega}{\Omega_i} \ll 1, \qquad \beta \doteq \frac{nT}{B^2/8\pi} \sim 1, \qquad \frac{U_e}{v_{Ti}} \sim \frac{U_i}{v_{Ti}} \sim 1. \tag{1}$$

(The assumption $\rho_i/L \ll 1$ is universal, others vary among different formulations of MHD.) Then, the "ion skip depth" $\delta_i \doteq c/\omega_{pi}$ and the "Alfvén speed", defined as

$$\delta_i \doteq \frac{c}{\omega_{pi}}, \qquad V_{\mathrm{A}} \doteq \frac{B}{\sqrt{4\pi\varrho_m}} \tag{2}$$

(here $\varrho_m \approx nm_i$ is the plasma mass density), satisfy

$$\frac{\delta_i}{\rho_i} \sim \frac{c}{\sqrt{4\pi ne^2/m_i}} \frac{eB/m_i c}{v_{Ti}} \sim \frac{1}{\sqrt{\beta}} \sim 1, \qquad \frac{V_{\mathrm{A}}}{v_{Ti}} \sim \frac{B}{\sqrt{4\pi n m_i v_{Ti}^2}} \sim \frac{1}{\sqrt{\beta}} \sim 1. \tag{3}$$

[1] In fact, MHD is not limited to plasmas. It is also commonly used to describe dynamics of liquid metals, like Earth's liquid iron, for example in the context of planetary dynamo. MHD predictions are also tested experimentally using liquid metals in laboratory.

Due to (1), this also yields

$$\frac{U_e}{V_\mathrm{A}} \sim \frac{U_i}{V_\mathrm{A}} \sim 1, \qquad \frac{V_\mathrm{A}}{c} \sim \frac{v_{Ti}}{c} \ll 1, \tag{4}$$

which leads to the following corollary. Since (1) justifies the guiding-center approximation, one can assume that the current density $\boldsymbol{j}$ is *at least* as large as the the polarization-current density $\boldsymbol{j}_\mathrm{pol} \sim (nm_ic^2/B^2)\omega E$. Then, $\boldsymbol{j}$ is much larger than the displacement current:

$$\frac{|c^{-1}\partial_t \boldsymbol{E}|}{|(4\pi/c)\boldsymbol{j}|} \lesssim \frac{\omega E}{(4\pi n m_i c^2/B^2)\omega E} = \frac{V_\mathrm{A}^2}{c^2} \ll 1. \tag{5}$$

(This is related to the fact established in the first homework: the low-frequency transverse susceptibility of cold magnetized plasma is $\chi_\perp = c^2/V_\mathrm{A}^2 \gg 1$.) Accordingly, Ampere's law can be used in a simplified form

$$\nabla \times \boldsymbol{B} = \frac{4\pi}{c}\boldsymbol{j}. \tag{6}$$

Unlike the arguments that led us here,[2] equation (6) is fundamental for MHD. It also yields the following estimate for the current:

$$j \sim \frac{cB}{4\pi L} \sim \epsilon \frac{cB}{4\pi\rho_i} \sim \epsilon \frac{cB}{4\pi\delta_i}\frac{\delta_i}{\rho_i} \sim \epsilon \frac{\omega_{pi}B}{4\pi}\frac{\delta_i}{\rho_i} \sim \epsilon n e V_\mathrm{A}\frac{\delta_i}{\rho_i} \sim \epsilon n e V_\mathrm{A}. \tag{7}$$

Using this and (4), one finds that electrons and ions have approximately the same fluid velocity:

$$\frac{|\boldsymbol{U}_e - \boldsymbol{U}_i|}{U_{e,i}} \sim \frac{j}{neU_{e,i}} \sim \epsilon \frac{V_\mathrm{A}}{U_{e,i}} \sim \epsilon \ll 1. \tag{8}$$

1.2 Electron physics and quasineutrality

We also need estimates for $n_e - n_i$ and E. For that, consider the electron momentum equation,

$$m_e n_e \frac{\mathrm{d}\boldsymbol{U}_e}{\mathrm{d}t} = -en_e\left(\boldsymbol{E} + \frac{\boldsymbol{U}_e}{c}\times\boldsymbol{B}\right) - \nabla\cdot\boldsymbol{P}_e + \boldsymbol{R}_{ei}. \tag{9}$$

Here, $\mathrm{d}/\mathrm{d}t = \partial_t + \boldsymbol{U}_e\cdot\nabla$ is the convective derivative whose two terms are generally comparable:

$$\frac{\boldsymbol{U}_e\cdot\nabla}{\partial_t} \sim \frac{v_{Ti}}{\omega L} \sim \left(\frac{\Omega_i}{\omega}\right)\left(\frac{\rho_i}{L}\right) \sim 1. \tag{10}$$

In the collisionless regime that we are interested in, the Braginskii formula for $\boldsymbol{R}_{ei}$ is not justified. In particular, the ∇T term is not to be expected, especially along $\boldsymbol{B}$, because transport is nonlocal. Still, one can expect that the friction term can be reasonably modeled as follows:

$$\boldsymbol{R}_{ei} = m_e n_e \nu_{ei}(\boldsymbol{U}_i - \boldsymbol{U}_e) = \frac{m_e\nu_{ei}}{e}\boldsymbol{j} \equiv \frac{en_e\boldsymbol{j}}{\sigma}, \qquad \sigma \doteq \frac{n_e e^2}{m_e\nu_{ei}}. \tag{11}$$

[2] Different authors arrive at (6) using different arguments, depending on a model. For example, requiring $V_\mathrm{A}/c \ll 1$ instead of $\beta \sim 1$ is enough.

(Note that σ looks suspiciously like the conductivity that we encountered earlier; hence the notation. We will discuss this later.) Basically, one can consider (11) as the first-order Taylor expansion in $\boldsymbol{U}_i - \boldsymbol{U}_e$ with *some* coefficient ν_{ei} that plays the role of the effective e–i collision rate. Strictly speaking, it is a matrix coefficient, but since we do not have a first-principle model for it anyway, let us treat it as a scalar for simplicity. This ν_{ei} may or may not be close to Bragniskii's ν_{ei} (it is often not), so consider (11) as a phenomenological closure. We will only assume that ν_{ei} is sufficiently small, specifically, such that $\epsilon\nu_{ei}/\Omega_e \ll 1$.

Let us now search for the dominant balance in (9). First of all, note that, relative to the magnetic term $en_e\boldsymbol{U}_e \times \boldsymbol{B}/c \sim enV_\mathrm{A}B/c$, the other terms in this equation scale as follows:

$$\frac{m_e n_e \,\mathrm{d}_t \boldsymbol{U}_e}{enV_\mathrm{A}B/c} \sim \frac{m_e n\omega V_\mathrm{A}}{enV_\mathrm{A}B/c} \sim \frac{\omega}{\Omega_e} \sim \epsilon\,\frac{m_e}{m_i} \ll 1, \tag{12}$$

$$\frac{\nabla\cdot\boldsymbol{P}_e}{enV_\mathrm{A}B/c} \sim \frac{nT/L}{enV_\mathrm{A}B/c} \sim \frac{cT}{\rho_i eV_\mathrm{A}B}\frac{\rho_i}{L} \sim \frac{v_{Ti}^2}{\rho_i V_\mathrm{A}\Omega_i}\frac{\rho_i}{L} \sim \frac{v_{Ti}}{V_\mathrm{A}}\frac{\rho_i}{L} \sim \frac{\rho_i}{L} \sim \epsilon \ll 1, \tag{13}$$

$$\frac{\boldsymbol{R}_{ei}}{enV_\mathrm{A}B/c} \sim \frac{m_e n\nu_{ei}\epsilon V_\mathrm{A}}{enV_\mathrm{A}B/c} \sim \frac{m_e\nu_{ei}\epsilon}{eB/c} \sim \epsilon\,\frac{\nu_{ei}}{\Omega_e} \ll 1. \tag{14}$$

Since all these ratios are small, the inertia, the pressure, and the friction cannot balance the magnetic Lorentz force in (9). We can conclude then that the magnetic Lorentz force is balanced by the only remaining term, the electric force. This leads to the following estimate:

$$E \sim \frac{V_\mathrm{A}}{c}\,B \sim \frac{v_{Ti}}{c}\,B. \tag{15}$$

(As a side remark: combined with (7), this readily shows that $j \sim j_\mathrm{pol}$.) Let us substitute (15) to Gauss's law to assess how much n_e can deviate from n_i:

$$\frac{n_e - n_i}{n} \sim \frac{E}{4\pi neL} \sim \frac{v_{Ti}B}{4\pi cneL} \sim \frac{m_i v_{Ti}}{4\pi ne^2L}\frac{eB}{m_i c} \sim \frac{T}{4\pi ne^2L}\frac{\Omega_i}{v_{Ti}}$$
$$\sim \frac{\lambda_D^2}{\rho_i L} \sim \epsilon\,\frac{\lambda_D^2}{\rho_i^2} \sim \epsilon\,\frac{\Omega_i^2}{\omega_{pi}^2} \sim \epsilon\,\frac{V_\mathrm{A}^2}{c^2} \ll 1. \tag{16}$$

Hence, we can assume $n_e \approx n_i \equiv n$. Also, let us introduce the center-of-mass velocity

$$\boldsymbol{U} \doteq \frac{n_e m_e \boldsymbol{U}_e + n_i m_i \boldsymbol{U}_i}{n_e m_e + n_i m_i} = \boldsymbol{U}_i + \mathcal{O}\left(\frac{m_e}{m_i}\right). \tag{17}$$

Using this and (8), the electron velocity can expressed as

$$\boldsymbol{U}_e = \boldsymbol{U}_i - (\boldsymbol{U}_i - \boldsymbol{U}_e) \approx \boldsymbol{U} - \boldsymbol{j}/ne = \boldsymbol{U} + \mathcal{O}(\epsilon). \tag{18}$$

Then, the magnetic Lorentz force in (9) can be expressed as follows:

$$\boldsymbol{U}_e \times \boldsymbol{B}/c = \boldsymbol{U} \times \boldsymbol{B}/c - \boldsymbol{j} \times \boldsymbol{B}/nec. \tag{19}$$

where the ratio of the second term of the right-hand side is $\sim \epsilon$ times the first one.

2 Main equations

2.1 Ohm's law

The inertia term in (9) is the smallest one and can be dropped. That and (19) allows one to rewrite (9) as

$$\boldsymbol{E} + \frac{\boldsymbol{U}}{c} \times \boldsymbol{B} = \frac{\boldsymbol{j} \times \boldsymbol{B}}{nec} - \frac{\nabla \cdot \boldsymbol{P}_e}{en} + \frac{\boldsymbol{j}}{\sigma}, \tag{20a}$$

where *all terms on the right-hand side are small.* (Accordingly, the expression on the left is also small but only because the two order-one components of the Lorentz force almost compensate each other.) Which of them should we keep? It depends on the physics that we seek to retain. To make a choice, let us rewrite this equation as follows:

$$\frac{\boldsymbol{j}}{\sigma} = \boldsymbol{E}' - \underbrace{\frac{\boldsymbol{j} \times \boldsymbol{B}}{nec}}_{\text{“Hall term”}} + \underbrace{\frac{\nabla \cdot \boldsymbol{P}_e}{en}}_{\text{“pressure term”}}, \qquad \boldsymbol{E}' \doteq \boldsymbol{E} + \frac{\boldsymbol{U}}{c} \times \boldsymbol{B}. \tag{20b}$$

In the absence of the Hall and pressure terms, one would obtain $\boldsymbol{j} = \sigma \boldsymbol{E}'$. Since $\boldsymbol{E}'$ is the electric field in the frame moving with the flow, one can recognize this as Ohm's law in the moving frame, with σ as the conductivity.[3] For this reason, (20) is called *generalized Ohm's law.* Below, we neglect the Hall term and the pressure term, because they only slightly modify Hamiltonian aspects of fluid dynamics. The term $\boldsymbol{j}/\sigma$, albeit also small, will be retained because it introduces something special: dissipation. This leads to the resistive Ohm's law

$$\boldsymbol{j} = \sigma(\boldsymbol{E} + \boldsymbol{U} \times \boldsymbol{B}/c) \tag{21}$$

and, as a limit at $\sigma \to \infty$, the ideal Ohm's law

$$\boldsymbol{E} + \boldsymbol{U} \times \boldsymbol{B}/c = 0. \tag{22}$$

Equation (21) is a part of resistive MHD, and (22) is a part of ideal MHD. Both of these will serve as equations for $\boldsymbol{E}$.

2.2 Momentum equation

Let us now consider the electron and ion momentum equations together,

$$m_e n_e \frac{\mathrm{d}\boldsymbol{U}_e}{\mathrm{d}t} \approx -e n_e \left(\boldsymbol{E} + \frac{\boldsymbol{U}_e}{c} \times \boldsymbol{B} \right) - \nabla \cdot \boldsymbol{P}_e + \boldsymbol{R}_{ei}, \tag{23a}$$

$$m_i n_i \frac{\mathrm{d}\boldsymbol{U}_i}{\mathrm{d}t} = +e n_i \left(\boldsymbol{E} + \frac{\boldsymbol{U}_i}{c} \times \boldsymbol{B} \right) - \nabla \cdot \boldsymbol{P}_i + \boldsymbol{R}_{ie}, \tag{23b}$$

[3] Strictly speaking, the current density is Galilean invariant only up to $\varrho \boldsymbol{U}$. However, as seen from (16), the charge density ϱ scales as $ne\epsilon V_A^2/c^2$, so $\varrho U/j \sim V_A^2/c^2 \ll 1$; i.e. $\varrho \boldsymbol{U}$ is negligible in this context.

and add them to each other. The friction forces cancel out by Newton's third law. Also, up to the mass ratio, the left-hand side is simply $\varrho_m \boldsymbol{U}$. Hence, one obtains

$$\varrho_m \frac{\mathrm{d}\boldsymbol{U}}{\mathrm{d}t} = e(n_i - n_e)\,\boldsymbol{E} + \frac{1}{c}\,\boldsymbol{j} \times \boldsymbol{B} - \nabla \cdot \boldsymbol{P}, \tag{24}$$

where $\boldsymbol{P} = \boldsymbol{P}_e + \boldsymbol{P}_i$. Let us compare the electric and magnetic terms:

$$\frac{|e(n_i - n_e)\,\boldsymbol{E}|}{|\boldsymbol{j} \times \boldsymbol{B}/c|} \sim \frac{n_i - n_e}{n}\,\frac{E}{V_\mathrm{A} B/c}\,\frac{enV_\mathrm{A}}{j} \sim \epsilon\,\frac{n_i - n_e}{n} \sim \epsilon^2\,\frac{V_\mathrm{A}^2}{c^2} \ll 1. \tag{25}$$

Then, the electric force can be neglected and one arrives at the following equation for $\boldsymbol{U}$:

$$\varrho_m \frac{\mathrm{d}\boldsymbol{U}}{\mathrm{d}t} = \frac{1}{c}\,\boldsymbol{j} \times \boldsymbol{B} - \nabla \cdot \boldsymbol{P}. \tag{26}$$

Using Ampere's law (6), one also finds that

$$\frac{1}{c}\,\boldsymbol{j} \times \boldsymbol{B} = \frac{1}{4\pi}\,(\nabla \times \boldsymbol{B}) \times \boldsymbol{B} = \frac{(\boldsymbol{B} \cdot \nabla)\boldsymbol{B}}{4\pi} - \frac{\nabla B^2}{8\pi}. \tag{27}$$

(One can recognize the right-hand side of (27) as the divergence of the the Maxwell stress tensor, where the electric-field contribution is negligible due to $E \sim (V_\mathrm{A}/c)B \ll B$.) Then, (26) can also be written in the following equivalent form:

$$\varrho_m \frac{\mathrm{d}\boldsymbol{U}}{\mathrm{d}t} = \frac{(\boldsymbol{B} \cdot \nabla)\boldsymbol{B}}{4\pi} - \nabla \cdot \left(\frac{B^2}{8\pi}\,\mathsf{I} + \boldsymbol{P}\right). \tag{28}$$

The first term, $(\boldsymbol{B} \cdot \nabla)\boldsymbol{B}/4\pi$ includes a force caused by the magnetic-field curvature $(\boldsymbol{b} \cdot \nabla)\boldsymbol{b}$, so it is loosely attributed to the tension of magnetic field lines. The second term, $-\nabla B^2/8\pi$, acts like a pressure force, so $B^2/8\pi$ is called magnetic pressure. The total pressure is given by $\boldsymbol{P} + (B^2/8\pi)\mathsf{I}$, and the contribution of $\boldsymbol{P}$ to this total pressure compared to $B^2/8\pi$ is $\sim \beta$.

2.3 Closure

At small β, or in the cold plasma limit, the kinetic pressure $\boldsymbol{P}$ can be neglected in (28). This is not an irrelevant model, as we will see later, but it is not the most interesting one either. More generally, we are lacking a constitutive equation, or a closure.

A rigorous closure is possible when collisions are entirely negligible. Then, in the appropriate limits, one can use the CGL double-adiabatic laws that we discussed earlier. The advantage of this approach is that the CGL equations actually follow from a proper asymptotic theory. The downside is that one has to distinguish $P_\parallel$ and $P_\perp$, which complicates the model.

This problem disappears in collisional closures: since they allow for Maxwellization, the pressure can be treated as a scalar. Assuming the energy exchange between electrons and ions is fast enough, i.e. $(m_e/m_i)\nu_{ei} \gg \omega$, one also has $T_e \approx T_i \equiv T$ and can use a single temperature equation, which is obtained by adding the temperature equations for electrons and ions:

$$\frac{3}{2}\,n_p \frac{\mathrm{d}T}{\mathrm{d}t} + P \nabla \cdot \boldsymbol{U} = S - \nabla \cdot \boldsymbol{Q} - 2\,\boldsymbol{\Pi} : \nabla \boldsymbol{U}, \qquad P = n_p T, \tag{29}$$

where $n_p \doteq n_e + n_i \approx 2n$ is the density of all plasma particles combined.[4] The problem then is how to calculate the right-hand side of (29). One can do this rigorously by using the Braginskii equations, but this requires $\lambda_{\rm mfp}/L \ll 1$. Unfortunately, the assumption of small $\lambda_{\rm mfp}/L$ is not even remotely satisfied for typical plasma-physics applications of MHD. Usually, this problem is handled by saying that the theory "mostly works anyway" and strong collisionality "almost" does not matter (Freidberg, 2014). But one is on shaky grounds here, because it remains unclear why, if truly unimportant, strong collisionality has to be assumed in the first place.

These difficulties are not merely technical, they reflect a real problem: in the absence of a small parameter, collisionless dynamics parallel to magnetic field is inherently kinetic, so there is no reason to assume *a priori* that a rigorous closure is possible. Fortunately, dynamics perpendicular to magnetic field is less dependent on the mean free path. Although the Braginskii equations are formally derived for small $\lambda_{\rm mfp}/L$, the perpendicular transport is primarily determined by ρ_i/L, which is legitimately small in MHD. Then, one can expect Braginskii's results to hold, at least qualitatively, for the transverse dynamics. This gives

$$\frac{\Pi_\times}{P} \sim \frac{U}{\Omega_i L} \sim \frac{v_{Ti}}{\Omega_i L} \sim \epsilon \ll 1, \qquad \frac{\Pi_\perp}{P} \sim \frac{\nu_{ii}}{\Omega_i}\frac{\Pi_\times}{P} \sim \epsilon\frac{\nu_{ii}}{\Omega_i} \ll 1, \tag{30}$$

$$Q_\times \sim \frac{P v_{Ti}^2}{\Omega_i}, \qquad Q_\perp \sim \frac{\nu_{ii}}{\Omega_i}\frac{P v_{Ti}^2}{m\Omega_i}. \tag{31}$$

To the extent the parallel dynamics can be ignored, these also ensure that

$$\frac{\boldsymbol{\Pi}_\times : \nabla \boldsymbol{U}}{P v_{Ti}/L} \sim \epsilon, \qquad \frac{\boldsymbol{\Pi}_\perp : \nabla \boldsymbol{U}}{P v_{Ti}/L} \sim \epsilon\frac{\nu_{ii}}{\Omega_i}, \qquad \frac{\nabla\cdot\boldsymbol{Q}_\times}{P v_{Ti}/L} \sim \epsilon, \qquad \frac{\boldsymbol{Q}_\perp}{P v_{Ti}/L} \sim \epsilon\frac{\nu_{ii}}{\Omega_i}. \tag{32}$$

In other words, as expected, the pressure can be treated as a scalar, $\boldsymbol{\Pi}$ is mostly ignorable, and so is the heat flux. The remaining term $S = S_{ei} + S_{ie}$ is the Joule heating:

$$S = \boldsymbol{R}_{ei}\cdot(\boldsymbol{U}_i - \boldsymbol{U}_e) = \frac{en}{\sigma}\boldsymbol{j}\cdot(\boldsymbol{U}_i - \boldsymbol{U}_e) = \frac{j^2}{\sigma}, \tag{33}$$

$$\frac{S}{P v_{Ti}/L} \sim \frac{m_e\nu_{ei}}{e^2 n}\frac{(\epsilon n e V_{\rm A})^2}{nT v_{Ti}/L} \sim \frac{m_e}{m_i}\frac{\nu_{ei}}{\Omega_i}\frac{\epsilon^2\Omega_i}{v_{Ti}/L} \sim \epsilon\frac{m_e}{m_i}\frac{\nu_{ei}}{\Omega_i} \sim \epsilon\sqrt{\frac{m_e}{m_i}}\frac{\nu_{ii}}{\Omega_i}. \tag{34}$$

To the extent that $\mathcal{O}(\epsilon)$ and smaller terms are negligible, one can neglect the entire right-hand side of (29) and arrive at adiabatic equation of state,

$$\frac{3}{2}\frac{{\rm d}T}{{\rm d}t} + T\nabla\cdot\boldsymbol{U} = 0, \qquad \text{or} \qquad \frac{\rm d}{{\rm d}t}\left(\frac{T}{n^{2/3}}\right) = 0, \tag{35}$$

where we used $\nabla\cdot\boldsymbol{U} = -{\rm d}_t \ln n$. This equation, which is just an entropy-conservation law, is considered a part of the ideal-MHD model, where it is usually presented in the equivalent form

$$\frac{\rm d}{{\rm d}t}\left(\frac{P}{\varrho_m^{5/3}}\right) = 0. \tag{36}$$

[4] Disturbingly, some authors choose to redefine the temperature as $T = T_e/2 = T_i/2$ to avoid introducing n_p or an additional factor of two in (29).

However, keep in mind: for weakly collisional plasmas such as fusion plasmas, this is not a rigorous result but rather a minimal conservative model with isotropic pressure.[5]

Now let us try to retain collisions. Since our previous arguments have not been particularly rigorous, it would be nice to avoid relying on the expressions for $\boldsymbol{Q}$ and $\boldsymbol{\Pi}$ and do something simple instead. A common approach in this case is to entirely neglect $\boldsymbol{Q}$ and $\boldsymbol{\Pi}$ but keep S to capture the fact that collisions can heat the plasma. This leads to the following equation:

$$2n\left(\frac{3}{2}\frac{\mathrm{d}T}{\mathrm{d}t}+T\nabla\cdot\boldsymbol{U}\right)=\frac{j^2}{\sigma}, \tag{37}$$

which is a part of resistive MHD. One can also reformulate it in terms of the pressure as

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{P}{\varrho_m^{5/3}}\right)=\frac{2j^2}{3\sigma\varrho_m^{5/3}}. \tag{38}$$

Note, however, that the omitted terms are actually much larger (!) than j^2/σ, both because $m_e/m_i \ll 1$ and because ν_{ii}/Ω_i is typically small too. This means that (37) is not rigorously justified and should be considered only as minimal model that captures Ohmic heating.

2.4 Continuity equation

The presence of the mass density ϱ_m in the momentum equation and the closures necessitates complementing the theory with an equation for ϱ_m. Since $\varrho_m \approx m_i n_i$, it can be just a rescaled continuity equation, which can be interpreted as mass conservation:

$$\partial_t\varrho_m+\nabla\cdot(\varrho_m\boldsymbol{U})=0. \tag{39}$$

2.5 Equation for the magnetic field

To close the system, we also need an equation for the magnetic field. For that, one can combine Faraday's lay with Ohm's law and use $\nabla\cdot\boldsymbol{B}=0$:

$$\begin{aligned}\frac{\partial\boldsymbol{B}}{\partial t}=-c\nabla\times\boldsymbol{E}&=-c\nabla\times\left(-\frac{\boldsymbol{U}}{c}\times\boldsymbol{B}+\frac{\boldsymbol{j}}{\sigma}\right)\\&=\nabla\times(\boldsymbol{U}\times\boldsymbol{B})-\frac{c}{\sigma}\nabla\times\frac{c\nabla\times\boldsymbol{B}}{4\pi}\\&=\nabla\times(\boldsymbol{U}\times\boldsymbol{B})+\frac{c^2}{4\pi\sigma}\nabla^2\boldsymbol{B},\end{aligned} \tag{40}$$

where we assumed, as commonly done in practice, that σ is a constant. Note that (40) is a diffusion equation. It can be written compactly as

$$\partial_t\boldsymbol{B}=\nabla\times(\boldsymbol{U}\times\boldsymbol{B})+\eta\nabla^2\boldsymbol{B}, \tag{41}$$

[5] Another popular conservative closure is the assumption of plasma incompressibility, $\nabla\cdot\boldsymbol{U}=0$. As a complement to (28), this equation can be used to calculate P. (Incompressibility does not mean that P is negligible!) However, this leads to omission of aspects of plasma dynamics that can be important depending on a problem.

where the "magnetic diffusivity" η is given by[6]

$$\eta \doteq \frac{c^2}{4\pi\sigma} = \nu_{ei}\,\frac{m_e c^2}{4\pi n e^2} = \nu_{ei}\delta_e^2, \tag{42}$$

and $\delta_e \doteq c/\omega_{pe}$ is the electron skin depth. In the ideal-MHD limit, where $\eta \to 0$, (41) reduces to

$$\partial_t \boldsymbol{B} = \nabla \times (\boldsymbol{U} \times \boldsymbol{B}). \tag{43}$$

2.6 Summary

Let us summarize. MHD is a single-fluid theory based on the following assumptions:

$$\epsilon \doteq \frac{\rho_i}{L} \sim \frac{\omega}{\Omega_i} \ll 1, \qquad \beta \doteq \frac{nT}{B^2/8\pi} \sim 1, \qquad \frac{U_e}{v_{Ti}} \sim \frac{U_i}{v_{Ti}} \sim 1, \tag{44}$$

plus $(m_e/m_i)\nu_{ei} \gg \omega$ to justify $T_e \approx T_i$. The complete set of resistive-MHD equations is[7]

$$\partial_t \varrho_m + \nabla \cdot (\varrho_m \boldsymbol{U}) = 0, \tag{45a}$$

$$\varrho_m \left(\partial_t + \boldsymbol{U} \cdot \nabla\right) \boldsymbol{U} = \boldsymbol{j} \times \boldsymbol{B}/c - \nabla P, \tag{45b}$$

$$\left(\partial_t + \boldsymbol{U} \cdot \nabla\right) \left(P/\varrho_m^{5/3}\right) = 2j^2/3\sigma\varrho_m^{5/3}, \tag{45c}$$

$$\partial_t \boldsymbol{B} = \nabla \times (\boldsymbol{U} \times \boldsymbol{B}) + \eta \nabla^2 \boldsymbol{B}, \tag{45d}$$

$$\boldsymbol{j} = c\boldsymbol{\nabla} \times \boldsymbol{B}/4\pi, \tag{45e}$$

$$\nabla \cdot \boldsymbol{B} = 0, \tag{45f}$$

where $\eta = c^2/4\pi\sigma$. Optionally, Ohm's law $\boldsymbol{j} = \sigma(\boldsymbol{E} + \boldsymbol{U} \times \boldsymbol{B}/c)$ can be included to determine the electric field. One can also substitute (45e) directly into (45b) and (45c) to eliminate $\boldsymbol{j}$ and work only with $(\rho, \boldsymbol{U}, P, \boldsymbol{B})$. In particular, (45b) can be expressed as

$$\varrho_m \left(\frac{\partial}{\partial t} + \boldsymbol{U} \cdot \nabla\right) \boldsymbol{U} = \frac{(\boldsymbol{B} \cdot \nabla)\boldsymbol{B}}{4\pi} - \nabla\left(\frac{B^2}{8\pi} + P\right). \tag{45g}$$

The ideal-MHD limit is obtained by taking $\sigma \to \infty$. Then, the right-hand side of (45c) and $\eta\nabla^2\boldsymbol{B}$ in (45d) vanish and Ohm's law becomes $\boldsymbol{E} + \boldsymbol{U} \times \boldsymbol{B}/c = 0$. Ideal MHD is particularly popular due to its simplicity and the fact that, as can be shown, it has Hamiltonian structure.

Other formulations of MHD, including two-fluid models, also exist and use assumptions different from (44), but this is a story for a separate course.

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[6] It is also common in MHD to use η as a notation for the resistivity, i.e. σ^{-1}. We do not introduce resistivity in these notes specifically to prevent this unfortunate clash of notation. Even in SI units, the magnetic diffusivity differs from the resistivity, by a coefficient.

[7] Memorizing the resistive term in (45c) is optional, but make sure to memorize the rest, including (44).

3 Alfvén waves

Magnetized plasmas within the MHD approximation support very special electromagnetic waves that are only found in magnetized conducting media. They were predicted by Alfvén in 1942 (Fig. 1). He received the 1970 Nobel Prize in Physics for this discovery (the only Nobel Prize ever awarded for work in plasma physics!), specifically, "for fundamental work and discoveries in magneto-hydrodynamics with fruitful applications in different parts of plasma physics".

These "Alfvén waves" are derived as follows. Let us assume the ideal-MHD limit and small β, so that the kinetic pressure is negligible. Let us also assume that the plasma is homogeneous and is at rest in a homogeneous magnetic field $\boldsymbol{B}_0 = B_0 \boldsymbol{b}_0$. Next, let us consider a small perturbation to the plasma velocity and the magnetic field,

$$\boldsymbol{U}(t,\boldsymbol{x}) = 0 + \tilde{\boldsymbol{U}}(t,\boldsymbol{x}), \qquad \boldsymbol{B}(t,\boldsymbol{x}) = \boldsymbol{B}_0 + \tilde{\boldsymbol{B}}(t,\boldsymbol{x}). \tag{46}$$

Then, the linearized momentum equation (45g) and the field equation (45d) are as follows:

$$\varrho_m \frac{\partial \tilde{\boldsymbol{U}}}{\partial t} = \frac{(\boldsymbol{B}_0\cdot\nabla)\tilde{\boldsymbol{B}}}{4\pi} - \nabla\left(\frac{\boldsymbol{B}_0\cdot\tilde{\boldsymbol{B}}}{4\pi}\right), \qquad \frac{\partial \tilde{\boldsymbol{B}}}{\partial t} = \nabla\times(\tilde{\boldsymbol{U}}\times\boldsymbol{B}_0), \tag{47}$$

where ϱ_m is now the *unperturbed* density. (The actual density is perturbed but there is no need for a separate notation since the density perturbation does not enter the linearized equations of interest.) Assuming the notation $\tilde{\boldsymbol{b}} \doteq \tilde{\boldsymbol{B}}/B_0$, one can readily rewrite these as follows:

$$\partial_t \tilde{\boldsymbol{U}} = V_\mathrm{A}^2(\partial_\parallel \tilde{\boldsymbol{b}} - \nabla \tilde{b}_\parallel), \qquad \partial_t \tilde{\boldsymbol{b}} = \nabla\times(\tilde{\boldsymbol{U}}\times\boldsymbol{b}_0), \tag{48}$$

where $_\parallel$ denotes the direction parallel to $\boldsymbol{B}_0$. Let us take the time derivative of the second equation and substitute the first one therein. This yields

$$\partial_t^2 \tilde{\boldsymbol{b}} = V_\mathrm{A}^2 \nabla\times((\partial_\parallel \tilde{\boldsymbol{b}} - \nabla\tilde{b}_\parallel)\times\boldsymbol{b}_0). \tag{49}$$

Let us assume $\tilde{\boldsymbol{b}} \propto \mathrm{e}^{-\mathrm{i}\omega t + \mathrm{i}\boldsymbol{k}\cdot\boldsymbol{x}}$, whence $\boldsymbol{k}\cdot\tilde{\boldsymbol{b}} = 0$ due to (45f). Then, one obtains

$$\begin{aligned}(\omega/V_\mathrm{A})^2\tilde{\boldsymbol{b}} &= \boldsymbol{k}\times((k_\parallel\tilde{\boldsymbol{b}} - \boldsymbol{k}\tilde{b}_\parallel)\times\boldsymbol{b}_0)\\ &= (k_\parallel\tilde{\boldsymbol{b}} - \boldsymbol{k}\tilde{b}_\parallel)k_\parallel - \boldsymbol{b}_0(k_\parallel\boldsymbol{k}\cdot\tilde{\boldsymbol{b}} - k^2\tilde{b}_\parallel)\\ &= k_\parallel^2\tilde{\boldsymbol{b}} - \boldsymbol{k}k_\parallel\tilde{b}_\parallel + k^2\boldsymbol{b}_0\tilde{b}_\parallel\\ &= k_\parallel^2\tilde{\boldsymbol{b}} - \boldsymbol{k}_\perp k_\parallel\tilde{b}_\parallel + k_\perp^2\boldsymbol{b}_0\tilde{b}_\parallel.\end{aligned} \tag{50}$$

Let us choose axes such that $\boldsymbol{b}_0$ is along the z-axis and $\boldsymbol{k}_\perp$ is along the x-axis. Then, the above equation can be written as $\boldsymbol{D}\tilde{\boldsymbol{b}} = 0$, where $\boldsymbol{D}$ is a matrix given by

$$\boldsymbol{D} = \begin{pmatrix} (\omega/V_\mathrm{A})^2 - k_\parallel^2 & 0 & k_\perp k_\parallel \\ 0 & (\omega/V_\mathrm{A})^2 - k_\parallel^2 & 0 \\ 0 & 0 & (\omega/V_\mathrm{A})^2 - k^2 \end{pmatrix}. \tag{51}$$

This matrix is disturbingly asymmetric, but this is because we introduced a constraint $\boldsymbol{k}\cdot\tilde{\boldsymbol{b}} = 0$. The unconstrained equation differs by $\boldsymbol{b}_0 k_\parallel(\boldsymbol{k}\cdot\tilde{\boldsymbol{b}}) = \boldsymbol{b}_0(k_\perp k_\parallel\tilde{b}_x + k_\parallel^2\tilde{b}_z)$, so it leads to

$$\boldsymbol{D} = \begin{pmatrix} (\omega/V_\mathrm{A})^2 - k_\parallel^2 & 0 & k_\perp k_\parallel \\ 0 & (\omega/V_\mathrm{A})^2 - k_\parallel^2 & 0 \\ k_\perp k_\parallel & 0 & (\omega/V_\mathrm{A})^2 - k_\perp^2 \end{pmatrix}. \tag{52}$$

In either case, by requiring that (50) has a nontrivial solution, i.e. $\det\boldsymbol{D} = 0$, one obtains two possible solutions for ω^2. Let us discuss them in detail.

Existence of Electromagnetic-Hydrodynamic Waves

If a conducting liquid is placed in a constant magnetic field, every motion of the liquid gives rise to an E.M.F. which produces electric currents. Owing to the magnetic field, these currents give mechanical forces which change the state of motion of the liquid. Thus a kind of combined electromagnetic-hydrodynamic wave is produced which, so far as I know, has as yet attracted no attention.

The phenomenon may be described by the electrodynamic equations

$$\text{rot } H = \frac{4\pi}{c} i$$

$$\text{rot } E = -\frac{1}{c}\frac{dB}{dt}$$

$$B = \mu H$$

$$i = \sigma(E + \frac{v}{c} \times B);$$

together with the hydrodynamic equation

$$\partial \frac{dv}{dt} = \frac{1}{c}(i \times B) - \text{grad } p,$$

where σ is the electric conductivity, μ the permeability, ∂ the mass density of the liquid, i the electric current, v the velocity of the liquid, and p the pressure.

Consider the simple case when $\sigma = \infty$, $\mu = 1$ and the imposed constant magnetic field H_0 is homogeneous and parallel to the z-axis. In order to study a plane wave we assume that all variables depend upon the time t and z only. If the velocity v is parallel to the x-axis, the current i is parallel to the y-axis and produces a variable magnetic field H' in the x-direction. By elementary calculation we obtain

$$\frac{d^2H'}{dz^2} = \frac{4\pi\partial}{H_0^2}\frac{d^2H'}{dt^2},$$

which means a wave in the direction of the z-axis with the velocity

$$V = \frac{H_0}{\sqrt{4\pi\partial}}.$$

Waves of this sort may be of importance in solar physics. As the sun has a general magnetic field, and as solar matter is a good conductor, the conditions for the existence of electromagnetic-hydrodynamic waves are satisfied. If in a region of the sun we have $H_0 = 15$ gauss and $\partial = 0{\cdot}005$ gm. cm.$^{-3}$, the velocity of the waves amounts to

$$V \sim 60 \text{ cm. sec.}^{-1}.$$

This is about the velocity with which the sunspot zone moves towards the equator during the sunspot cycle. The above values of H_0 and ∂ refer to a distance of about 10^{10} cm. below the solar surface where the original cause of the sunspots may be found. Thus it is possible that the sunspots are associated with a magnetic and mechanical disturbance proceeding as an electromagnetic-hydrodynamic wave.

The matter is further discussed in a paper which will appear in *Arkiv för matematik, astronomi och fysik*.

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Aug. 24.

Figure 1: The *entire* paper by Alfvén (1942) that eventually brought him a Nobel prize.

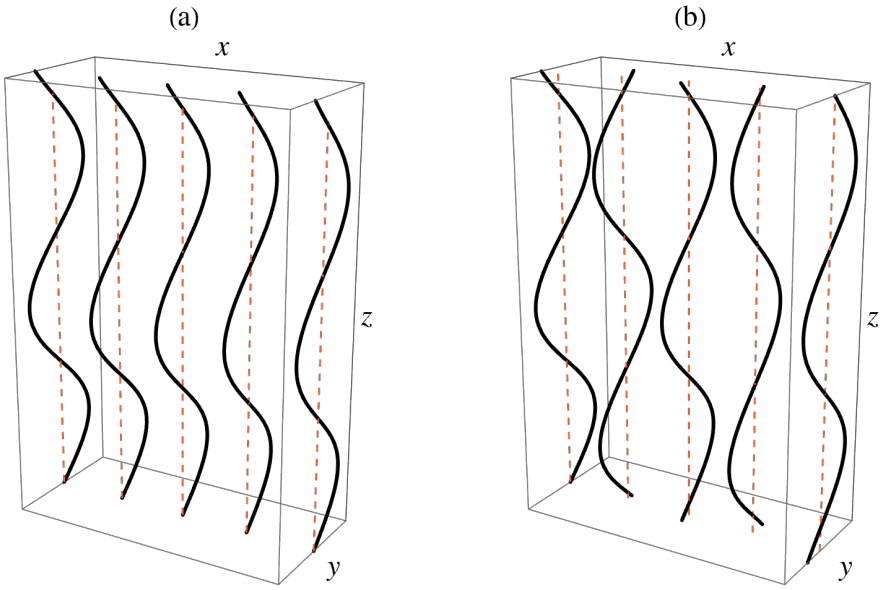


Figure 2: Magnetic field lines in a shear Alfvén wave: (a) $k_\perp = 0$; (b) nonzero $k_\perp$. Red dashed are the corresponding unperturbed field lines.

3.0.1 Shear Aflvén waves

The first solution corresponds to

$$\omega^2 = k_\parallel^2 V_\mathrm{A}^2, \qquad \begin{pmatrix} 0 & 0 & \blacksquare \\ 0 & 0 & 0 \\ 0 & 0 & \blacksquare \end{pmatrix} \begin{pmatrix} \tilde{b}_x \\ \tilde{b}_y \\ \tilde{b}_z \end{pmatrix} = 0, \tag{53}$$

where we used (52) and ▪ denotes nonzero elements. Clearly, this mode has $\tilde{b}_z = 0$, and $\tilde{b}_x = 0$ too, due to $0 = \nabla \cdot \tilde{\boldsymbol{b}} = k_\perp \tilde{b}_x + k_\parallel \tilde{b}_z$. However, $\tilde{b}_y$ can be nonzero. From (48), it is seen that its only nonzero velocity component then is $\tilde{U}_y$, so

$$\nabla \cdot \tilde{\boldsymbol{U}} = k_\perp \tilde{U}_x + k_\parallel \tilde{U}_z = 0, \qquad \partial_t \tilde{n} = -n_0 \nabla \cdot \tilde{\boldsymbol{U}} = 0, \tag{54}$$

i.e. such waves do not perturb the plasma density. They are called *shear Alfvén waves*, and they are driven by magnetic tension. Think about magnetic field lines as guitar strings: a small transverse perturbation makes the string vibrates perpendicularly, causing a wave that propagates along the field [Fig. 2(a)]. As we will discuss later, the fact that ω depends only on $k_\parallel$ ensures that the wave energy propagates only along $\boldsymbol{B}_0$. However, this does *not* mean that such a wave cannot have nonzero $k_\perp$. In this case, the wave becomes essentially three-dimensional, because $\boldsymbol{k}$ has nonzero x- and z-components and $\tilde{\boldsymbol{b}}$ is along the y-axis [Fig. 2(b)].

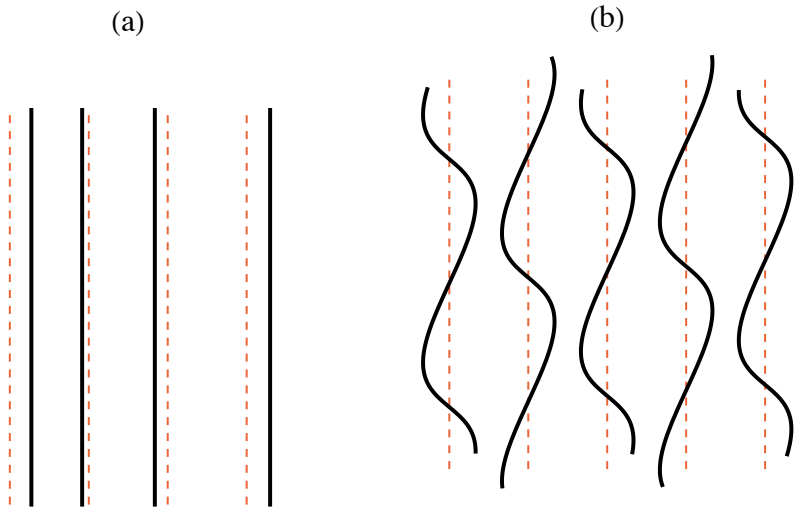


Figure 3: Magnetic field lines in a compressional Alfvén wave: (a) $k_\parallel = 0$; (b) nonzero $k_\parallel$. Red dashed are the corresponding unperturbed field lines.

3.0.2 Compressional Aflvén waves

The second solution corresponds to

$$\omega^2 = k^2 V_\mathrm{A}^2, \qquad \begin{pmatrix} \blacksquare & 0 & \blacksquare \\ 0 & \blacksquare & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \tilde{b}_x \\ \tilde{b}_y \\ \tilde{b}_z \end{pmatrix} = 0. \tag{55}$$

Clearly, such waves have $\tilde{b}_y = 0$ but possibly nonzero $\tilde{b}_z$ and $\tilde{b}_x = -(k_\parallel/k_\perp)\tilde{b}_z$. Nonzero $\tilde{b}_z$ implies compression of magnetic field lines, which is also associated with density perturbations. Such waves are called *compressional Alfvén waves* (Fig. 3).

Quiz

- What is MHD, in broad strokes?
- What are the assumptions that MHD is based on?
- What is generalized Ohm's law?
- Explain, without details, how the generalized Ohm's law is obtained.
- What is the ideal Ohm's law?
- What is resistive MHD?
- What is ideal MHD?
- What is the magnetic diffusivity?

- What is the complete list of resistive MHD equations?
- What is the complete list of ideal MHD equations?
- What is the dispersion relation and basic physics of shear Alfvén waves?
- What is the dispersion relation and basic physics of compressional Alfvén waves?

References

Alfvén, H. 1942 Existence of electromagnetic-hydrodynamic waves. *Nature* **150** (3805), 405.

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